GG-O-635A
March 29, 1977
SUPERSEDING
Fed. Spec. GG-O-635
May 31, 1960

FEDERAL SPECIFICATION

OPTICAL FLATS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers optical flats used in laboratories, and in gage and inspection departments. They are for use with a monochromatic light source to determine flatness, parallelism or size of gages, by interpretation of light interference patterns and bands.

1.1.1 Federal specification coverage. Federal specifications do not include all types and classes of the commodity which are commercially available, but include only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and classes. Optical flats covered by this specification shall be of the following types and classes as specified (see 6.1).

Type I - Reference flats.

Class 1 - Single surface.
Class 2 - Double surface.

Type II - Master flats.

Class 1 - Single surface.
Class 2 - Double surface.

Type III - Working flats.

Class 1 - Single surface.
Class 2 - Double surface.
Class 3 - Double surface parallel.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-591	-	Boxes, Shipping, Fiberboard, Wood-Cleated.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-640	-	Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-B-665	-	Boxes, Paperboard, Metal Edged and Components.
PPP-B-676	-	Boxes, Setup.
PPP-B-1364	-	Boxes, Shipping, Corrugated Fiberboard, High Strength, Weather Resistant, Double-Wall.
PPP-T-76	-	Tape, Packaging Paper, (For Carton Sealing).
PPP-T-97	-	Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

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Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

Military Specifications:

MIL-P-116 - Preservation-Packaging, Methods of.
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

American Society for Testing and Materials (ASTM) Standard:

E 380 - Metric Practice Guide.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. The material used in the optical flats shall be either fused quartz or a borosilicate glass (pyrex) as specified (see 6.1).

3.1.1 Fused quartz. Unless otherwise specified (see 6.1), optical flats shall be furnished of relatively clear and colorless fused quartz, free from extraneous matter. The quartz shall be finely annealed and shall be free of manufacturing defects which would affect the functional characteristics of the flat. Occasional bubbles, not to exceed 0.05 inch (1.27 mm) in diameter and not exceeding two in number in any cubic inch, may be permitted provided they do not interfere with the viewing performance. The coefficient of expansion of the fused quartz shall not exceed 0.32 millionths per degree fahrenheit (0.58 millionths per degree centigrade).

3.1.2 Borosilicate glass (pyrex). When borosilicate glass or pyrex is specified (see 6.1), the material for the optical flats shall be furnished of relatively clear and colorless borosilicate glass, and shall be free from extraneous matter. The borosilicate shall be finely annealed and shall be free of manufacturing defects which would affect the functional characteristics of the flat. Occasional bubbles, not to exceed 0.05 inch (1.27 mm) in diameter and not exceeding two in number in any cubic inch, may be permitted provided they do not interfere with the viewing performance. The coefficient of expansion of the borosilicate glass shall not exceed 2 millionths per degree fahrenheit (3.6 millionths per degree centigrade).

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3.2 Design. Flats of all types and classes shall be cylindrical in shape and shall be designed to produce accurate straight interference bands from edge to edge of the working surface when suitably illuminated by a monochromatic light and placed in contact with a known flat surface and normally viewed through the optical flat. The edges of the flats shall be beveled to a 45-degree angle and shall be smooth, either fine ground or medium polished to avoid chipping at the edges. The beveled edge distance shall be 0.008 to 0.012 times the nominal diameter of the particular flat. The working surfaces of the flats shall be optically polished to a flat and regular surface without zones or aberrations of greater vertical height or depth than one third the total amount of the flatness tolerance. The working surfaces shall be free from measurable surface roughness, pebbling, greyness or any scratches visible to the naked eye. The face opposite the working surface of the class 1 flats shall be polished to at least window clarity for freedom from ordinary difficulties in seeing into the flat.

3.3 Accuracies. Optical flats shall conform to the tolerances specified for the respective types and classes. The fall-off of the fringes in excess of 1/8 inch (3.2 mm) of the edge of the working surface shall not exceed the tolerance specified for flatness. The working surfaces shall be flat within the tolerance specified to a distance not in excess of 1/8 inch (3.2 mm) from outer edges. The flatness tolerance shall be the total distance measured at a constant standard temperature of 70° plus or minus 5° (21 plus or minus 3°C) along a line parallel to the axes of the flat from the highest point on the surface to the lowest point on the surface. A calibration certificate shall be furnished with each optical flat. Each certificate shall indicate the deviation in millionths of an inch from true flatness, if any, and whether concave or convex.

3.3.1 Small flats. For optical flats of 4 inches (102 mm) diameter or less or of areas of 12 sq inches (7742 sq mm) or less, the flatness tolerances shall be as follows:

(a) Type I, reference flats. The flatness tolerance on working surfaces of type I, reference optical flats shall not exceed plus or minus 0.000001 inches (0.000025 mm).

(b) Type II, master flats. The flatness tolerance on working surfaces of type II, master optical flats shall not exceed plus or minus 0.000002 inches (0.000051 mm).

(c) Type III, working flats. The flatness tolerance on working surfaces of type III, working optical flats shall not exceed plus or minus 0.000004 inches (0.000102 mm).

3.3.2 Large flats. For optical flats of sizes larger than 4 inches (102 mm) diameter or of areas larger than 12 sq inches (7742 sq mm), the flatness tolerances shall be not more than 1-1/2 times the tolerances as specified in 3.3.1.

3.4 Classes.

3.4.1 Class 1. Class 1 optical flats shall have ground and polished ends and one surface shall be flat within the tolerance accuracy specified for the applicable type. The working surface shall be indicated by a single-ended arrow located on the cylindrical edge.

3.4.2 Class 2. Class 2 optical flats shall have ground and polished ends and both surfaces shall be flat within the tolerance accuracy specified for the applicable type. The working surfaces shall be indicated by a double-ended arrow located on the cylindrical edge.

3.4.3 Class 3. Class 3 optical parallel flats shall have both opposing working surfaces ground and polished. Both of these working surfaces shall have a flatness within plus or minus 0.000004 inches (0.000102 mm) and shall be parallel within 0.000004 inches (0.000102 mm) for a linear distance that equals the diameter of the flat. These parallel optical flats shall be indicated with a double-ended arrow, along with the thickness in hundred-thousandths of an inch, marked along the cylindrical edge.

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3.5 Working surface indications and dimensions. The working surfaces of all types and classes of optical flats shall be indicated by arrows located on the cylindrical edges.

3.5.1 Classes 1 and 2. Flatness tolerances of classes 1 and 2 optical flats shall be as specified in 3.4.1 and 3.4.2, as applicable. Unless otherwise specified (see 6.1), dimensions of optical flats of classes 1 and 2 of all types shall be as specified in table I. The opposing flat surfaces shall be sufficiently parallel to prevent distortion of indicating lines. Unless otherwise specified (see 6.1), optical flats shall be supplied as a single optical flat enclosed with a case.

TABLE I. Dimensions of classes 1 and 2 of types I, II, and II optical flats

Diameter, Minimum (Inches)		Thickness, Minimum (Inches)	
1	(25.4 mm)	1/4	(6.3 mm)
1-1/4	(31.7 mm)	1/4	(6.3 mm)
2	(50.8 mm)	1/2	(12.7 mm)
3	(76.2 mm)	9/16	(14.2 mm)
4	(101.6 mm)	11/16	(17.4 mm)
5	(127 mm)	7/8	(22.2 mm)
6	(152.4 mm)	1	(25.4 mm)
8	(203.2 mm)	1-5/16	(33.3 mm)
10	(254 mm)	1-11/16	(42.8 mm)
12	(304.8 mm)	2	(50.8 mm)

3.5.2 Class 3 of type III. Flatness and parallelism tolerance shall be as specified in 3.4.3. Unless otherwise specified (see 6.1), dimensions of optical flats of class 3, type III shall be as specified in table II. Class 3, type III parallel flats shall be supplied in sets of two or four, as specified (see 6.1).

3.6 Case. A case, of the manufacturer's choice with the exceptions specified herein, shall be provided with each flat or set of flats required. The case shall be designed to prevent the optical flats from contacting one another and to provide maximum protection of the flats during handling and storage. The case shall be provided with hinges and a latch of rugged construction. All metal parts shall be fabricated of corrosion-resistant material. An exception to the corrosion-resistant coating on the retention case is that it shall not include cadmium plating. The case shall be lined with a soft felt or equally protective material to protect the finished surfaces of the optical flats from any scratches or abrasions while being stored or transported in the cases.

TABLE II. Dimensions of parallel flats, type III, class 3

Diameter, Minimum (Inches)		Thickness (Inches) + 0.0001 inches (+ 0.00254 mm)	
1	(25.4 mm)	0.50000	(12.70000 mm)
1	(25.4 mm)	0.50630	(12.86002 mm)
1	(25.4 mm)	0.51250	(13.01750 mm)
1	(25.4 mm)	0.51880	(13.17752 mm)

3.7 Coating on class 1 flats. The coating on class 1 optical flats shall be durable enough to withstand the intended usage and shall not be subject to peeling or flaking off. This coating shall be of a material that will improve the capability or the intended optical characteristics of the optical flat. This coating shall only be required on an, as specified, basis and shall only be applicable to fused quartz material (see 6.1).

3.8 Center hole. The center hole in optical flats shall be concentric within 0.0005 inches, (0.0127 mm) total indicator reading, with the outer perimeter, and its diameter shall not exceed one-third of the optical flat's overall diameter. A center hole will only be acceptable on an, as specified, basis (see 6.1).

3.9 Marking. In addition to the marking requirements previously specified, each flat shall be marked to show its accuracy in millionths of an inch and with the manufacturer's name or trademark of such know character that the source of manufacture may readily be determined.

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3.10 Workmanship. Each optical flat shall be ground and polished to the extent that all surface defects that would impair usefulness and serviceability are removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1, the contractor is responsible for insuring that materials and components used were manufacturered, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table III.

TABLE III. Sampling data

Category	Sample Unit	Inspection Level	Acceptable Quality Level	AQL Expressed In Terms Of	Ref.
Visual and dimensional examination	1 each	I	1.5	% Defective	4.3
Testing	1 each	I	1.5	% Defective	4.4
Preparation for Delivery One Container	S-2	4.0		Defects per Hundred Units	4.5

4.3 Examination.

4.3.1 Visual and dimensionsal examination. Each sample unit shall be examined for any nonconformance in design, material, finish, workmanship, and marking.

4.4 Testing. Each sample unit shall be tested in accordance with 4.4.1 and 4.4.2.

4.4.1 Flatness test. The working surface of the flat shall be subjected to a flatness test. The test shall be conducted on flats under suitably controlled uniform temperature. Light interference patterns of a monochromatic light source shall indicate the working surface of the flat to be accurate within the tolerance specified for the specific type and class. Flats larger than 6 inches in diameter shall be tested by being supported vertically and tested against an accurately made spherical mirror using the Foucault knife edge test or they shall be tested for flatness by any approved method of known accuracy.

4.4.2 Parallelism test. Optical flats requiring parallel surfaces (type III, class 3) shall be tested for parallelism of the working surfaces. Using an electrolimit or air gage comparator, graduated in 0.000001 inch, readings shall be taken at five points on the optical parallel; one point shall be the approximate center of the optical parallel and the other four shall be spaced 90° apart on a circumference having a radius approximately 90 percent times the radius of the optical parallel. The maximum deviation between the five readings is the total error of parallelism and shall not exceed the tolerance of parallelism specified in 3.4.3. A three-button anvil is essential for supporting the optical parallel to prevent air from being trapped between the optical parallel and a flat anvil. Other approved methods of known accuracy may also be used for this test.

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4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing, and markings comply with the requirements in section 5. Defects shall be scored in accordance with table IV. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot.

TABLE IV. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. Optical flats shall be preserved and packaged in accordance with method III of MIL-P-116. Rough handling test shall be required.

5.1.1.1 Intermediate containers. Unit quantities in an intermediate container shall be as specified (see 6.1). Intermediate containers shall conform to PPP-B-566, PPP-B-676, PPP-B-665, or PPP-B-636, at the option of the contractor. Box closures shall conform to the applicable box specification or the appendix thereto when contained therein. The gross weight of paperboard boxes shall not exceed 10 pounds and fiber boxes shall not exceed 20 pounds.

5.1.2 Level C. Optical flats shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing.

5.2.1 Level A. Optical flats, packaged as specified (see 6.1), shall be packed in boxes conforming to PPP-B-585, class 2; PPP-B-591, class II; PPP-B-601, overseas type; PPP-B-621, class 2; PPP-B-636, class weather-resistant or class waterproof and water vapor resistant; PPP-B-640, class 2; or PPP-B-1364, at the option of the supplier. Shipping containers shall have caseliners conforming to MIL-L-10547. Caseliners shall be closed and sealed in accordance with the appendix to MIL-L-10547. Caseliners for boxes conforming to PPP-B-636 may be omitted provided all joints and corners of the boxes are sealed with minimum 1-1/2 inch (38 mm) wide tape conforming to PPP-T-76. Boxes shall be closed and strapped in accordance with the applicable box specification or appendix thereto, except fiber boxes may be banded with tape conforming to type IV of PPP-T-97. The gross weight of wood or wood cleated boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitation of the applicable box specification. Intermediate fiber boxes need not be overpacked if they conform to weather-resistance of waterproof and water vapor resistance of PPP-B-636, closed, sealed and banded as specified herein, when used as the shipping container.

5.2.2 Level B. Optical flats, packaged as specified (see 6.1), shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with the appendix to PPP-B-636. Intermediate containers conforming to PPP-B-636, closed as specified herein, may be used as shipping containers without overpacking.

5.2.3 Level C. Optical flats, preserved and packaged as specified (see 6.1) shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the Uniform Freight Classification rules and the National Motor Freight Classification rules.

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5.3 Marking.

5.3.1 Civil agencies. Each unit, intermediate and shipping container shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order on herein specified interior and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, diameter, and thickness required (see 1.2.1, table I, table II, 3.5.1, and 3.5.2).
(c) If fused quartz is not required (see 3.1, 3.1.1, and 3.1.2).
(d) Quantity of flats required per case, if different (see 3.5.1, and 3.5.2).
(e) Coating on class 1 flats, if required (see 3.7).
(f) A center hole, if required of if acceptable (see 3.8).
(g) Levels of preservation, packaging, packing, and marking required (see section 5).

6.2 Metric conversions. Conversions from English to Metric system of measurement, as shown in this specification, involves use of ASTM E 380. Metric equivalents are shown in parentheses.

MILITARY INTERESTS:

Custodians:

Air Force - 99
Army - MU

Reviewers:

Army - CD

Users:

Army - AV

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

Preparing Activity:

Air Force - 99

PROJECT NUMBER: 5210-0221

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein.
Price 30 cents each.

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DOCUMENT IDENTIFIER AND TITLE

CG-O-635A Optical Flats

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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